

From Benchmark Accuracy to Deployment Reliability: A Stress-Test Framework for Flow-Based Network Intrusion Detection

Ihsen Alaya
FAIR-ML-CYBER project
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Abstract

Machine-learning-based network intrusion detection systems often report near-perfect performance on benchmark datasets such as CICIDS2017. Such results are frequently obtained under random train/test splits and insufficient controls for temporal, scenario, endpoint, rare-class, open-set and calibration effects. This paper presents FAIR-ML-CYBER, a reproducible stress-test framework for flow-based network intrusion detection. Using 2,438,052 CICIDS2017-like flows from 18 CSV files, we compare random stratified evaluation with temporal, latest-day, scenario and endpoint-pair holdouts. We further evaluate multiclass labels, rare attacks, unknown-family holdouts, calibration, abstention and explanation stability. Full-data binary experiments over three seeds show that random stratified evaluation is substantially more optimistic than deployment-oriented stress tests. HistGradientBoosting reaches a mean macro-F1 of 0.9977 under random stratification but drops to 0.2315 under temporal holdout, while LogisticRegression drops from 0.9357 to 0.5401. Advanced seed-42 analyses show severe multiclass degradation, rare-class failure under Web holdout, unstable open-set behavior and large calibration errors under temporal and latest-day shifts. A partial external validation on real CSE-CIC-IDS2018 CICFlowMeter CSVs shows the same random-vs-stress concern: on a 200,928-row stratified sample, HistGradientBoosting drops from 0.9224 random macro-F1 to 0.7408 under Web holdout, and LogisticRegression drops from 0.8618 to 0.6459. The results support a benchmark-validity contribution: high random-split scores do not imply deployment reliability. The study provides a tracked protocol, evidence artifacts and operational lessons for more rigorous NIDS evaluation.

Keywords: network intrusion detection; CICIDS2017; dataset bias; temporal validation; open-set recognition; calibration; explainability; reproducibility.

1 Introduction

Network Intrusion Detection Systems (NIDS) are a common target for machine learning research, yet the practical value of many reported results remains contested. A recurring problem is that benchmark performance can be extremely high under random splits while failing to reflect deployment conditions. In operational environments, future traffic may differ by time, service, endpoint, attack campaign and unknown attack family. A model that succeeds under an IID random split can therefore still be unreliable for a Security Operations Center (SOC).

This problem is particularly visible for CICIDS2017-like datasets. CICIDS2017 remains widely used because it offers labelled benign and malicious flows, multiple attack scenarios and feature-rich CSV outputs [13, 2]. At the same time, several studies have documented dataset-quality, flow-construction and labelling limitations [5, 10]. More broadly, NIDS dataset surveys

and pragmatic assessments of machine learning for NIDS warn that dataset choice, evaluation protocol and reproducibility strongly affect conclusions [9, 7, 14, 1].

This manuscript evaluates whether CICIDS2017-like evidence can support a strong scientific contribution if the paper is framed as a benchmark-validity and deployment-reliability study rather than as a new detector paper. The main experiments use the locally available CSV files and apply stress tests designed to approximate deployment shifts: temporal holdout, latest-day holdout, Web scenario holdout, endpoint-pair holdout and unknown-family open-set holdout. We also record calibration, abstention and explanation-stability behavior. To reduce the single-dataset risk, we add a partial external validation on real CSE-CIC-IDS2018 CICFlowMeter CSV files [3].

Contributions. The manuscript makes four practical contributions:

1. A reproducible evaluation framework for full-data flow-based NIDS experiments with dataset hashes, feature hashes, split hashes, Azure ML job provenance and tracked evidence artifacts.
2. A multi-seed binary stress-test analysis showing that random stratified performance can severely overstate deployment reliability.
3. A seed-42 advanced analysis covering multiclass labels, rare classes, unknown attack families, calibration, abstention and explanation stability.
4. A partial external validation on CSE-CIC-IDS2018, including binary stress tests, post-hoc calibration and anomaly-based open-set baselines.
5. A critical discussion of why these results should be interpreted as benchmark-validity evidence, not as a claim of universal intrusion-detection superiority.

2 Related Work

2.1 NIDS datasets and CICIDS2017

Sharafaldin et al. introduced CICIDS2017 as a labelled intrusion detection dataset with benign and malicious traffic scenarios [13]. The dataset has become a standard benchmark, but this popularity also makes it difficult to publish another paper that only reports high accuracy on the same data. Ring et al. survey NIDS datasets and emphasize the importance of dataset properties and quality criteria [9]. Goldschmidt and Chuda extend this discussion in a recent survey of public NIDS datasets, highlighting selection bias, dataset limitations and the risk of weak justification for dataset use [7].

2.2 Known limitations of CICIDS2017

Engelen et al. analyze CICIDS2017 and report issues in traffic generation, flow construction, feature extraction and labelling [5]. Rosay et al. further analyze CICIDS2017 and propose alternative flow extraction through LycoSTand/LYCOS-IDS2017 [10]. These studies motivate a cautious interpretation: high scores on CICIDS2017-like data are not automatically evidence of a deployable model.

2.3 Evaluation methodology and generalization

Sommer and Paxson argued early that applying machine learning to intrusion detection differs from closed-world classification because attacks are rare, environments drift and operational costs matter [14]. Apruzzese et al. similarly call for pragmatic NIDS evaluation that goes beyond benchmark outperformance [1]. Cantone et al. show that cross-dataset generalization

can degrade sharply even when within-dataset scores are high [4]. Our main evidence remains an intra-dataset stress-test study, complemented by a partial external check on CSE-CIC-IDS2018 rather than a full cross-dataset benchmark.

2.4 Feature standardization, uncertainty and explainability

Sarhan et al. argue for standard feature sets to improve comparability and generalizability across NIDS datasets [11, 12]. Related work on uncertainty and explainability motivates calibration, abstention and explanation-stability analyses [15, 8]. Open-set intrusion detection has also been studied through unknown-traffic detection and open-set recognition methods [6]. Our current open-set analysis is intentionally simple: it evaluates whether classifier uncertainty can rank unknown-family traffic. It is therefore a baseline diagnostic, not a complete open-set solution.

3 Dataset and Audit

The study uses 18 CSV files located in the project dataset directory. The verified audit reports 2,438,052 rows and a full dataset hash of `f51899df9bd60758`. The files resemble CICIDS2017 or a derived variant, with labels such as benign traffic, DoS, DDoS, PortScan, Botnet, brute-force attacks, Web attacks and Heartbleed. The experiments add a binary label for benign versus attack and preserve original labels for multiclass and rare-class evaluation.

The dataset is highly imbalanced. Rare labels such as Heartbleed and Web SQL Injection have very small support, which makes accuracy and weighted metrics insufficient. We therefore report macro-F1, per-class metrics, AUROC/AUPRC where appropriate, calibration metrics and selective-prediction metrics.

3.1 External validation dataset

We also downloaded five public CSE-CIC-IDS2018 CICFlowMeter CSV files from the official UNB/AWS source [3]. The audited subset contains 4,525,400 rows after removing repeated embedded header rows and has hash `414f7195ee1e137a`. The downloaded subset contains Benign, Bot, FTP-BruteForce, SSH-Bruteforce, Infiltration, Brute Force-Web, Brute Force-XSS and SQL Injection labels. Because benign traffic dominates, we generated a real-data stratified validation sample with 80,000 benign rows, up to 30,000 rows per attack label and all rare Web/XSS/SQL rows. The resulting sample contains 200,928 rows and has hash `d90aca82728746b3`. This sample is not synthetic. It is a capped subset of real CSE-CIC-IDS2018 rows. Endpoint-pair holdout is unavailable for this subset because the public processed CSV files do not include source and destination IP columns.

4 Methods

4.1 Feature tiers

The current full-data experiments focus on two non-identifying feature tiers:

- **no_identity**: excludes explicit identifiers such as flow id, IPs and timestamp.
- **deployment_safe**: excludes identity/context columns and keeps features that are more plausible for a SOC deployment.

The advanced analyses use **deployment_safe** only. Future work should extend the final manuscript with service/port and standardized NetFlow-like tiers.

The implementation excludes timestamp-derived fields, including **hour**, from non-identifying feature tiers. This correction is important for future final reruns because timestamp-derived context can encode scenario timing.

4.2 Splits

We evaluate five split protocols:

1. **Random stratified split**: conventional IID-like baseline.
2. **Temporal split**: train on earlier flows and test on later flows.
3. **Latest-day holdout**: hold out the latest day, 2017-07-07.
4. **Scenario holdout**: hold out the Web scenario.
5. **Endpoint-pair holdout**: split by endpoint-pair groups.

For open-set analysis, one attack family is excluded from training and evaluated as unknown at test time. The current unknown families are Web, Botnet, PortScan and DDoS.

4.3 Models

The verified full-data core experiments use LogisticRegression (LR) and HistGradientBoosting (HGB). RandomForest was validated in pilot experiments but was not executed in full-data core due to memory/artifact-risk constraints. LogisticRegression initially produced convergence warnings at `max_iter=500`. A seed-42 rerun with `max_iter=2000` completed 10/10 LR runs with zero convergence warnings.

4.4 Metrics

Binary metrics include macro-F1, balanced accuracy, MCC, AUROC, AUPRC, Brier score and Expected Calibration Error (ECE). Multiclass metrics include macro-F1 and weighted-F1. Rare-class analysis reports per-label precision, recall and F1. Open-set analysis reports unknown-family uncertainty AUROC/AUPRC, attack recall as attack and unknown recall at the 90th percentile uncertainty review threshold. Abstention reports coverage, review rate, selective error, selective macro-F1 and accepted attack recall. Explanation stability is measured as top-15 feature-overlap Jaccard against the random-stratified baseline.

4.5 Statistical analysis

For the full-data binary core, we compute paired random-versus-stress comparisons over the six available experimental units per model: three seeds crossed with two feature tiers. We report the mean macro-F1 delta, the stress/random ratio, bootstrap 95% confidence intervals over those experimental units and exact paired sign-flip p-values. These intervals quantify variation across seeds and feature tiers; they are not per-flow confidence intervals.

4.6 Reproducibility

Every reported full-data run records dataset hash, feature hash, split hash, seed, model, feature tier, split, runtime and artifact paths. Azure ML job metadata records Git commits and dirty flags. Evidence snapshots are tracked under the repository `evidence/` directory.

5 Results

5.1 Full-data binary core across three seeds

Table 1 reports the mean macro-F1 by split across seeds 42, 7 and 99, averaged over the two feature tiers. The results show a sharp gap between random stratified evaluation and deployment-oriented stress tests.

Table 1: Mean macro-F1 by split across seeds 42, 7 and 99. Values are averaged over no_identity and deployment_safe.

Split	HGB	LR
Random stratified	0.9977	0.9357
Endpoint-pair holdout	0.9924	0.8820
Temporal	0.2315	0.5401
Latest-day holdout	0.4090	0.6394
Scenario holdout Web	0.5207	0.4831

HGB is nearly perfect under random stratification but collapses under temporal holdout. LR is weaker under random stratification but more stable under temporal and latest-day shifts. Endpoint-pair holdout remains high for both models and should not be over-interpreted as proof of deployment robustness.

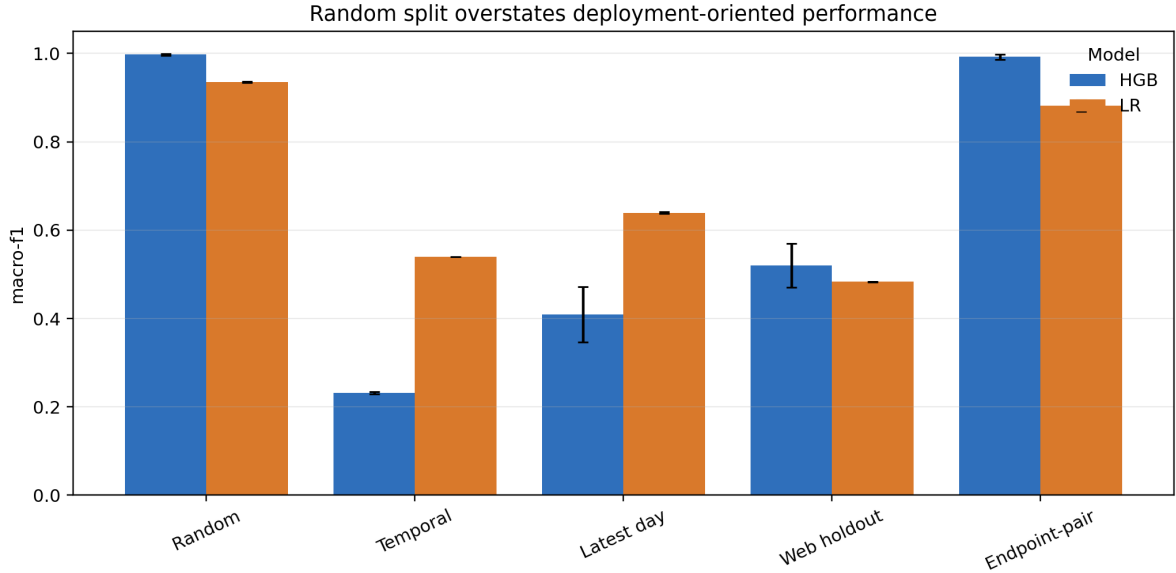


Figure 1: Full-data macro-F1 by split and model across seeds 42, 7 and 99. Bars show means over seeds and feature tiers; error bars show standard deviation over the same experimental units.

5.2 Cyber Transferability Score

We define a simple Cyber Transferability Score (CTS):

$$\text{CTS}(m, s) = \frac{\text{macroF1}(m, s)}{\text{macroF1}(m, \text{random})},$$

where m is the model and s is a stress-test split. Table 2 reports mean CTS across seeds.

These values quantify the main message: high random-split macro-F1 does not imply high transferability under time or scenario shifts.

5.3 Paired robustness statistics

Table 3 reports paired stress-minus-random deltas and stress/random ratios. All stress splits are worse than the paired random baseline for both models. The exact sign-flip p-value is 0.0313

Table 2: CTS macro-F1 mean and standard deviation across seeds.

Target split	Model	CTS mean	CTS std
Endpoint-pair holdout	HGB	0.9946	0.0048
Temporal	HGB	0.2321	0.0013
Latest-day holdout	HGB	0.4099	0.0341
Scenario holdout Web	HGB	0.5219	0.0387
Endpoint-pair holdout	LR	0.9426	0.0155
Temporal	LR	0.5772	0.0004
Latest-day holdout	LR	0.6834	0.0018
Scenario holdout Web	LR	0.5163	0.0004

in every row because all six paired differences have the same sign; with six pairs this is the smallest possible two-sided value. Standardized effect sizes are not shown in the table because very low inter-seed variance can make them numerically unstable.

Table 3: Paired random-vs-stress macro-F1 statistics across seeds and feature tiers. CI values are bootstrap 95% intervals over six paired experimental units.

Model	Stress split	Delta [95% CI]	Ratio [95% CI]	p
HGB	Temporal	-0.7662 [-0.7674, -0.7650]	0.2320 [0.2307, 0.2337]	0.0313
HGB	Latest-day	-0.5887 [-0.6281, -0.5486]	0.4099 [0.3697, 0.4505]	0.0313
HGB	Web holdout	-0.4770 [-0.4996, -0.4353]	0.5219 [0.4993, 0.5634]	0.0313
HGB	Endpoint-pair	-0.0054 [-0.0089, -0.0020]	0.9946 [0.9906, 0.9980]	0.0313
LR	Temporal	-0.3956 [-0.3964, -0.3947]	0.5772 [0.5767, 0.5777]	0.0313
LR	Latest-day	-0.2963 [-0.2980, -0.2947]	0.6834 [0.6817, 0.6851]	0.0313
LR	Web holdout	-0.4526 [-0.4534, -0.4516]	0.5163 [0.5157, 0.5170]	0.0313
LR	Endpoint-pair	-0.0537 [-0.0632, -0.0443]	0.9426 [0.9325, 0.9527]	0.0313

5.4 Binary calibration under stress

The seed-42 advanced run provides binary calibration metrics for the `deployment_safe` tier. Table 4 shows macro-F1 and ECE. The hardest shifts are not only lower-performing; they are poorly calibrated.

Table 4: Seed-42 binary macro-F1 and ECE for the `deployment_safe` feature tier.

Model	Split	Macro-F1	ECE
HGB	Random stratified	0.9960	0.0009
LR	Random stratified	0.9370	0.0345
HGB	Temporal	0.2296	0.7013
LR	Temporal	0.5395	0.3955
HGB	Latest-day holdout	0.3678	0.4180
LR	Latest-day holdout	0.6370	0.2528
HGB	Scenario holdout Web	0.6218	0.0093
LR	Scenario holdout Web	0.4833	0.0661
HGB	Endpoint-pair holdout	0.9917	0.0020
LR	Endpoint-pair holdout	0.8837	0.0584

Temporal ECE reaches 0.7013 for HGB and 0.3955 for LR, suggesting severe overconfidence

or miscalibration under shift.

5.5 Partial external validation on CSE-CIC-IDS2018

Table 5 reports the binary partial external validation on the 200,928-row CSE-CIC-IDS2018 stratified sample. The same qualitative concern appears outside the local CICIDS2017-like CSV files: random-split scores are higher than Web scenario holdout scores for all three tested models.

Table 5: Partial external validation on real CSE-CIC-IDS2018 CSVs. Macro-F1 on the 200,928-row stratified validation sample.

Model	Random	Temporal	Latest-day	Web holdout
LR	0.8618	0.6727	0.6604	0.6459
HGB	0.9224	0.8312	0.8204	0.7408
RF	0.9158	0.9284	0.9195	0.7193

Post-hoc calibration on this external sample is split-dependent. For LR, isotonic calibration reduces random-split ECE from 0.0395 to 0.0040 and temporal ECE from 0.3619 to 0.1533, but Web-holdout ECE worsens from 0.1693 to 0.2309. For HGB, isotonic calibration reduces random ECE from 0.0151 to 0.0029, but temporal ECE worsens from 0.0864 to 0.1310. Open-set scoring on a 50,000-row external subsample shows that Local Outlier Factor is much stronger than Isolation Forest for Web, Botnet and BruteForce unknown-family scoring, with AUROC values of 0.8990, 0.9574 and 0.9087 respectively. These results strengthen the conclusion that calibration and open-set behavior must be evaluated under each shift rather than assumed from random-split performance.

5.6 Multiclass collapse under shifted splits

Table 6 reports seed-42 multiclass macro-F1 and weighted-F1. Weighted-F1 can remain high even when macro-F1 collapses, especially in scenario and endpoint holdouts. This confirms that weighted aggregate metrics mask minority-class failures.

Table 6: Seed-42 multiclass results for the `deployment_safe` feature tier.

Model	Split	Macro-F1	Weighted-F1
HGB	Random stratified	0.7479	0.9901
LR	Random stratified	0.4487	0.9175
HGB	Temporal	0.0415	0.1360
LR	Temporal	0.0453	0.1621
HGB	Latest-day holdout	0.0523	0.4256
LR	Latest-day holdout	0.0546	0.4439
HGB	Scenario holdout Web	0.0762	0.9791
LR	Scenario holdout Web	0.0667	0.9225
HGB	Endpoint-pair holdout	0.4447	0.9923
LR	Endpoint-pair holdout	0.3044	0.9051

The Azure log emitted 16 sklearn warnings stating that predicted labels contained classes absent from the true labels during shifted multiclass/per-class evaluation. This is not a cosmetic warning; it indicates label-support shift and should be discussed as part of the failure mode.

5.7 Rare-class behavior

Rare-class results further expose the weakness of global metrics. In Web scenario holdout, all three Web rare classes have F1 equal to 0.0 for both LR and HGB despite nonzero test support. Table 7 reports the verified values.

Table 7: Rare Web-class F1 under Web scenario holdout.

Label	Support	LR F1	HGB F1
Web Brute Force	2734	0.0000	0.0000
Web SQL Injection	24	0.0000	0.0000
Web XSS	1358	0.0000	0.0000

This result should be framed as a controlled failure analysis. It does not show that the models solve rare attack detection; it shows that random or weighted metrics can hide rare-class failure.

5.8 Open-set unknown-family holdout

Table 8 reports unknown-family uncertainty metrics. Uncertainty behavior is model- and family-dependent. HGB uncertainty ranks unknown traffic well for all four families, whereas LR fails badly for DDoS uncertainty ranking despite high binary attack recall.

Table 8: Seed-42 open-set unknown-family results.

Unknown family	Model	Support	Uncertainty AUROC	Unknown recall at p90
Web	LR	4116	0.8697	0.0649
Web	HGB	4116	0.9353	0.6618
Botnet	LR	5508	0.7607	0.0005
Botnet	HGB	5508	0.9262	0.6899
PortScan	LR	161323	0.9501	0.2348
PortScan	HGB	161323	0.9954	0.4393
DDoS	LR	95733	0.2019	0.0545
DDoS	HGB	95733	0.9587	0.3570

This is a diagnostic baseline rather than a complete open-set method. A stronger final paper should add anomaly or open-set baselines.

5.9 Abstention and selective prediction

At approximately 90% coverage, selective macro-F1 remains poor on hard shifts. For example, HGB temporal selective macro-F1 is 0.2416 and accepted attack recall is 0.0000; LR temporal selective macro-F1 is 0.5931 with accepted attack recall 0.3975. At 50% coverage, LR improves on temporal and latest-day holdouts, but this requires sending half of the flows to review. The practical interpretation is therefore a coverage-risk tradeoff, not an automatic model improvement.

5.10 Explanation stability

Top-15 explanation-overlap Jaccard against the random-stratified baseline is much lower for HGB than LR under the hardest shifts. HGB temporal Jaccard is 0.0714, corresponding to only two overlapping top-15 features, whereas LR temporal Jaccard is 0.5789. This suggests

that HGB feature importance is more sensitive to split protocol. However, the current HGB permutation importance uses a sample size of 2000 and one repeat; a final journal version should increase repeats and add uncertainty estimates.

6 Discussion

The results support three conclusions.

First, random-split benchmark accuracy is not a reliable proxy for deployment reliability. HGB appears nearly perfect under random stratification, but its temporal macro-F1 and calibration collapse. LR is less impressive under random stratification but can be more stable under some shifts.

Second, binary evaluation is insufficient. Multiclass macro-F1, rare-class results and Web holdout results reveal failures that are hidden by weighted or binary metrics. This is particularly important for SOC deployment, where missing a rare but critical attack can be operationally unacceptable.

Third, uncertainty and explanations are not automatically robust. Open-set results vary strongly by model and unknown family, while explanation stability changes across split protocols. These behaviors should be evaluated as first-class outcomes, not as optional post-hoc figures.

7 Threats to Validity

External-validation scope. The main evidence still comes from one CICIDS2017-like dataset. The CSE-CIC-IDS2018 result is a partial validation using five public CSV files and a stratified real-data sample, not a full cross-dataset benchmark. The paper should therefore claim external consistency evidence, not universal generalization.

Known CICIDS2017 limitations. Prior studies report CICIDS2017 flow-construction and labelling issues. These issues motivate the study but also limit claims about real-world attack detection.

Advanced analyses are seed 42 only. The binary core is validated on seeds 42, 7 and 99. The advanced analyses are currently seed 42 only. Final claims about rare-class, open-set, calibration and explanation stability should be repeated on additional seeds if compute budget allows.

LogisticRegression convergence. LR convergence is fixed for seed 42 with `max_iter=2000`. Older seed 7 and seed 99 LR rows still contain warnings from `max_iter=500`. Corrected LR2000 reruns without the timestamp-derived `hour` feature have been submitted for seeds 42, 7 and 99, but their artifacts were not yet available at the time of this draft. Final LR multi-seed tables should use those convergence-clean reruns or explicitly report this limitation.

Open-set baseline strength. The main advanced open-set analysis uses classifier uncertainty. The external validation adds Isolation Forest and Local Outlier Factor, but these are still baseline anomaly detectors. A final version should compare additional open-set or anomaly approaches if scope permits.

Explanation precision. HGB explanation stability uses lightweight permutation importance. A final version should use more repeats and confidence intervals.

8 Conclusion

FAIR-ML-CYBER shows that CICIDS2017-like evidence can support a meaningful Q1-oriented contribution if the paper is framed as a rigorous evaluation and benchmark-validity study, and if its limits are tested against external data. The verified results show that random-split scores are overly optimistic, temporal and scenario shifts expose major transferability failures, calibration degrades under shift, rare classes are poorly detected, open-set behavior is inconsistent and explanation stability depends on the split protocol. The partial CSE-CIC-IDS2018 validation supports the same caution: random-split performance remains insufficient evidence of deployment reliability.

The manuscript should not claim a new state-of-the-art detector. Its strongest claim is methodological: NIDS evaluation must include leakage-aware stress tests, transferability metrics, rare-class analysis, calibration, abstention, open-set diagnostics and explanation stability before benchmark scores can be interpreted as deployment evidence.

Reproducibility Statement

The repository records Azure ML job names, Git commits, dataset hashes, feature hashes, split hashes, runtimes and evidence snapshots. The main verified artifacts are:

- `evidence/fullcore-mem-s42-001/`
- `evidence/fullcore-mem-s7-001/`
- `evidence/fullcore-mem-s99-001/`
- `evidence/fullcore-lr2000-s42-001/`
- `evidence/q1-stats-fullcore-multiseed-001/`
- `evidence/advanced-core-s42-001/`
- `evidence/cse-cic-ids2018-partial-s42-001/`

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